

Algebra 1  
Unit 9  
Lesson 10 Practice

Name Answer Key  
Date \_\_\_\_\_  
Period \_\_\_\_\_

1. Use the quadratic formula to solve  $x^2 - 14x + 66 = 4x$ .

$$\begin{aligned}x^2 - 18x + 66 &= 0 \\x &= \frac{18 \pm \sqrt{(-18)^2 - 4(1)(66)}}{2(1)} \\x &= \frac{18 \pm \sqrt{60}}{2} \\x &= 9 \pm \sqrt{15}\end{aligned}$$

2. Solve the equation,  $4x^2 + 32x + 60 = 0$ , by factoring.

$$\begin{aligned}4(x^2 + 8x + 15) &= 0 \\4(x + 5)(x + 3) &= 0 \\x + 5 &= 0 & x + 3 &= 0 \\x &= -5 & x &= -3\end{aligned}$$

3. Solve the equation,  $x^2 - 16x = -39$ , by completing the square.

$$\begin{aligned}x^2 - 16x + 64 &= -39 + 64 \\\sqrt{(x - 8)^2} &= \sqrt{25} \\x - 8 &= \pm 5 \\x &= 8 + 5 & x &= 8 - 5 \\x &= 13 & x &= 3\end{aligned}$$

4. Solve the equation,  $2x^2 - 2 = 3x$ , with the method of your choice.

$$\begin{aligned}2x^2 - 3x - 2 &= 0 \\(2x + 1)(x - 2) &= 0 \\x &= -\frac{1}{2} & x &= 2\end{aligned}$$

The equation  $H(x) = -16x^2 + 64x + 192$  models the height,  $H$ , in feet, of a toy rocket after  $x$  seconds that it is launched from the top of a 16-story building.

5. Solve  $-16x^2 + 64x + 192 = 0$  by factoring.

$$\begin{aligned}-16(x^2 - 4x - 12) &= 0 \\-16(x - 6)(x + 2) &= 0 \\x - 6 &= 0 & x + 2 &= 0 \\x &= 6 & x &= -2\end{aligned}$$

6. Interpret the solutions and determine if the solutions are viable.  
After 6 seconds and  $-2$  seconds, the toy rocket is on the ground.  
 $-2$  seconds is not viable because time cannot be negative.

A statistician for an outdoor company analyzes data for the years 2007 to 2014 to help inform the company about trends. The function  $R(x) = -14x^2 + 112x + 776$  models the number of people ages 6 to 17, in tens of thousands,  $R$ , who listed running as a hobby for  $x$  years since 2007.

7. Solve  $0 = -14x^2 + 112x + 776$  using the quadratic formula.

$$x = \frac{-112 \pm \sqrt{(112)^2 - 4(-14)(776)}}{2(-14)}$$

$$x = \frac{-112 \pm \sqrt{56,000}}{-28}$$

$$x = -4.45$$

$$x = 12.45$$

8. Interpret the solutions and determine if the solutions are viable.

12.45: In the year 2019, 0 people listed running as a hobby. Not viable, outside constraints for the domain.

-4.45: -4.45 years 0 people listed running as a hobby. Not viable – outside the domain

Several investors provided funds for a startup company to begin production of a new product. The company's revenue,  $R$ , in thousands of dollars, for the first year is modeled by the function  $R(x) = 4x^2 - 23x + 30$ , where  $x$  is the number of months since production started.

9. Solve  $0 = 4x^2 - 23x + 30$  using the method of your choice.

$$(4x - 15)(x - 2) = 0$$

$$4x - 15 = 0$$

$$x = 3.75$$

$$x - 2 = 0$$

$$x = 2$$

10. Interpret the solutions and determine if the solutions are viable.

2 months after production started, the company had \$0 in revenue.

3.75 months after production started, the company had \$0 in revenue.

Both viable solutions